

Introduction to Machine Learning & Deep Neural Networks

CS 401 | Lecture Series: Foundations of Artificial Intelligence

Presented by Prof. Catherine Hughes & AI Research Lab

The Three Core Paradigms of Machine Learning

1. **Supervised Learning:** Learning with ground-truth labeled pairs (X, Y).
 - Classification (Logistic Regression, Random Forests, SVM, ResNet)
 - Regression (Linear Regression, Gradient Boosted Trees, MLP)
2. **Unsupervised Learning:** Discovering latent structure without explicit targets.
 - Clustering (K-Means, DBSCAN, Hierarchical)
 - Dimensionality Reduction (PCA, t-SNE, UMAP, Autoencoders)
3. **Reinforcement Learning:** Agent maximizing expected cumulative reward via environment interaction (Markov Decision Processes, Q-Learning, PPO).

Deep Neural Networks: Backpropagation & Optimization

- **Forward Propagation:** Computes layer-wise representations $Z[l] = W[l]A[l-1] + b[l]$, $A[l] = g(Z[l])$.
- **Loss Functions:** Categorical Cross-Entropy, Mean Squared Error, Huber Loss.
- **Backpropagation:** Uses the chain rule of calculus to compute analytical gradients of loss with respect to all trainable parameters.
- **Modern Optimizers:** Stochastic Gradient Descent with Momentum, Adam, AdamW with cosine annealing learning rate schedules.